Proceedings of the Second International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS-2024)
IEEE Xplore Part Number: CFP24DN7-ART; ISBN: 979-8-3503-6841-3

2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS) | 979-8-3503-6841-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSSAS64001.2024.10760503

A Comparative Technique with Credit card based Fraud detection with Machine Learning and Quantum Algorithm

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***Abstract*— Quantum computing greatly benefit a wide range of fields, including the detection of credit card fraud, through its twin characteristics of improved computational power and efficiency. Quantum algorithms are used to enhance data analysis, pattern recognition and optimization properly with increased speed and efficiency. In the proposed model, we investigate how quantum machine learnings algorithms (QML) are used to spot credit card fraud using a data set that includes transactions from European cardholders. Various models such as neural network activation functions, epochs, and other factors employed in a hybrid quantum-classical method. The results from the best model employing a quantum hybrid technique, a conventional neural network, and an ML approach are compared. For equivalent prediction performance, we find that the quantum models train substantially more effectively than the conventional models. In order to implement Quantum algorithms, penny lane is used for enabling differentiable scripting of quantum computers that is cross-platform. Similar to how a neural network is trained, a quantum computer is trained. Overall, this in-depth investigation of the application of hybrid QML in credit card fraud detection shows significant promise.**

Keywords—Quantum machine learning, Quantum Twist, fraud detection, credit card detection, machine learning

I. INTRODUCTION

Over the past 10 years, the Internet has experienced considerable growth. Services like online bill payment systems, e-commerce, tap-and-pay systems, etc. have multiplied and expanded as a result. [1] The outcome is, assaults by credit card fraudsters on transactions are now more prevalent than ever. Only two of the security procedures in place to safeguard credit card transactions are tokenization and cardholder information protection. Though these methods often work, they cannot totally protect credit card operations against fraud.

When individuals use the term "credit card" it means that they refer to something that one has access to and normally allows for the purchase of goods or services to a certain credit limit, or withdrawals of cash in advance. Payment cards provide cardholders the benefit of time by enabling consumers to postpone repayment for a certain period of time by carrying the amount over to the next cycle of payments. Credit card fraud is a common problem. Without any hazards, a sizeable sum may

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be taken out secretly in a short length of time without the owner's awareness. Every fraudulent transaction is always made to appear real by the fraudsters, making it incredibly difficult to identify fraud. The development of Quantum Machine Learning (QML)[2] recently has drawn attention to quantum technology, making it difficult to predict whether QML would outperform conventional machine learning or not. The depth or consecutive gate operations are constrained because the real devices are noisy [3-5]. The information will also be lost when qubits lose their entanglement.

A few instances with encouraging outcomes and certain firms' dedication to the quantum machine learning path are already available on the market. One example of how a scoring model for risk assessment may be constructed is CaixaBank (a Spanish bank), which used the Pennylane quantum framework to design and test QML models [6].

The two main obstacles to even understanding a trustworthy result remain to be this output or input paradigm and the availability of good usable qubits. IBM achieved the threshold of 100+ [7], but it is still insufficient to apply sophisticated algorithms that, depending on the kind of issue to be solved, call for thousands or even millions of qubits [8]. various table text styles are provided. The formatter will need to create these components, incorporating the applicable criteria that follow. The proposed model implements quantum algorithm for credit card fraud detection and methodology.

The organization of the paper is as follows. They are in section II literature review is discussed. In Section III the proposed System architecture is discussed. In Section IV quantum machine learning algorithms which are implemented are discussed. In Section V the results and analysis are discussed. Finally the proposed model concludes and future work are discussed.

II. Literature Review

It [9] is challenging to show the present and prospective quantum advantage of QML approaches since the available quantum computers are noisy and have a limited number of qubits to test. This work demonstrates that we can improve the performance of quantum classifiers and their classical encoding. During data preparation stage by employing linear discriminant analysis (LDA). As a consequence, with the LDA technique, the Variational Quantum Algorithm (VQA) performs better than baseline conventional classifiers and displays enhanced balanced accuracy. The advantages are quantum computer may leverage classification results by utilizing only a few dimensions and can extract more useful information from traditional data. The disadvantages are in traditional ML, the study of PCA on unsupervised tasks is still being decided, and it does not provide information on which techniques perform better on particular types of data.

In [10] One of the greatest ways to predict frauds is to use the classifier with the higher rating score. As a result, a feedback mechanism is used to address the idea problem. The primary goal of the project is to create and implement a unique fraud detection algorithm for streaming transaction data in order to assess historical customer transaction data and extract behavioral patterns. The advantage of this work is it analysis historical customer spends and analysis the fraud with precision. The disadvantage of this technique is this Quantum methods are not used here which could provide a faster results compared to the traditional once.

In [11] The study and use of two strategies for creating a technique for identifying credit card fraud are covered in this article. hybrid quantum neural networks are used in the first one. Recently, quantum computers (QC) have started to enter the AI/ML space. The usage of hybrid quantum neural networks (QNN) and conventional neural networks has been applied to a variety of applications, including image processing and natural language processing. In the second strategy, topological data analysis is used (TDA). From the standpoint of noise reduction, identifying topological structure in the incoming data also becomes important. TDA's visualization capabilities may also be used to help identify credit card fraud. Here, TDA is applied using a mapper-based technique. The advantage of this technique is quantum algorithms are faster than normal algorithms. The disadvantage of this method is still in the research stage so there isn't much evidence to support the paper and the comparison with ml current model is not shown to compare the accuracy.

In [12] "Credit card fraud" is the term used to describe either the actual loss of a credit card or the loss of private credit card information. For identification, a variety of machine-learning techniques can be applied. This inquiry provides a number of methods for determining if transactions are false or authentic. The study utilizes a dataset for identifying fraudulent use of credit cards. The SMOTE approach was used for oversampling since the dataset was extremely unbalanced. After going through a feature selection process, the dataset was divided into training and test halves. Multilayer Perceptron, Logistic Regression, Naive Bayes, and Random Forest were the algorithms employed in the study. Findings demonstrate that each algorithm is capable of accurately detecting credit card fraud. The advantage of this technique Research shows that every single model can identify credit card theft with accuracy. The disadvantage of this model is they do not use quantum algorithms which makes the process highly efficient to implement

In [13] this study, Genetic algorithm (GA) is the technique for choosing features for machine learning (ML) depending on identifying fraudulent credit card transactions. The suggested detection engine employs the ML classifiers Artificial Neural Network (ANN), Decision Tree (DT), Logistic Regression (LR), and Random Forest (RF) after selecting the best features. Using a dataset created

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from European cardholders, the suggested credit card fraud detection engine's effectiveness is evaluated. The advantage of this technique is outcome proved our suggested strategy works better than current systems. The disadvantages of this techniques aren't enough datasets used to generalize the process; quantum computing can also be used to further make the process efficient.

In [14] At the nexus of quantum technology and artificial intelligence, quantum machine learning is a quickly expanding discipline. This study offers a dual overview of a number of important strategies that may progress the development of quantum technologies and the capability of artificial intelligence. Quantum-enhanced algorithms [19] are one of these methods, which strengthen cornerstone machine learning solutions by applying quantum software engineering to conventional information processing. In this context, we investigate how hybrid quantum-classical neural networks can enhance precision and generalization of models while consuming less processing power. The Advantage of this gives paper gives insight into converting physics quantum knowledge to software engineering. Further the proposed model has all the three models used like classical neural network, normal machine learning models and quantum machine learning models. The above literatures don't have all the functionalities together combined with quantum algorithms and less evidences are provided to implement this. From the literature the following gaps are identified:
The surge in digitization of financial transactions brought a tremendous spike to the volume of credit card deals all around the world. Although the introduction of digital financial systems has made it easier for people to transfer cash and have seamless transactions, credit card fraud has also increased the risks. Through this, the financial institutions and the consumers are facing major difficulties in security regarding phishing attacks. The traditional manner in which fraud detection is done, like setting up of rules or classical machine learning methods, faces an increase in fraud sophistication and dynamics and as a result are very weak. Therefore, it becomes necessary to investigate advanced methods that can make fraud detection systems more accurate, efficient, and robust. Quantum computing has become an emerging field that bases its principles on quantum mechanics. It provides a very promising path for finding the solution to these problems. Quantum Neural Networks (QNNs) that bring together the concepts of quantum computing and neural networks are believed to change the way the fraud detection area is approached. The special parallelism and better computing power that quantum computers have can give a strong boost to the efficiency of the neural networks to a great extent .Focus has been where state-of-the-art fraud detection applications could bring about through quantum neural networks, but it is still the fact that we are far from bridging the yawning research gaps of model development, data encoding, and realistic application. Existing QNN models are not made specifically for the high-dimensional and imbalanced condition of transaction datasets, which in turn hinders their performance. There are also no unified, efficient methods for quantum states of transaction data to be encoded, so this becomes a major issue for practical applications of QNNs. Moreover, the question of how much quantum hardware can be scaled for QNNs which will allow efficient fraud detection on the large-scale datasets against the white bench remains open without any empirical studies that confirm their functionality. It is crucial to solve these problems with focused research to be able to use QNNs for better and faster fraud detection

III. Proposed System Architecture

The normal machine learning algorithms are very slow compared to the quantum algorithms. The proposed System architecture is divided into three modules and 4 layers. They are data collection, feature extraction, training and testing and comparing the results.

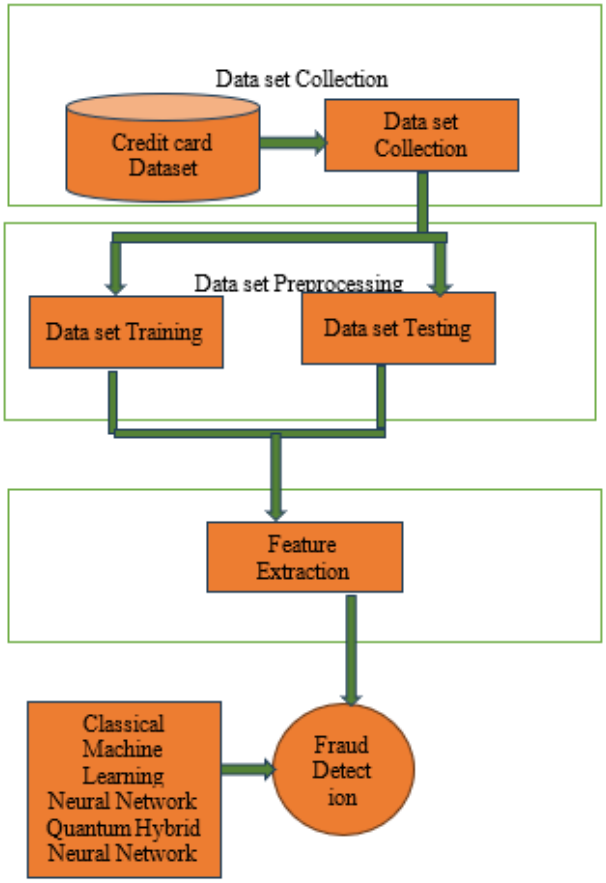

Fig 1. Proposed Four layer architecture for credit card fraud detection

The proposed work gets dataset from Kaggle where it contains transactions made by credit card holder by the year 2013 by European card holders. The dataset is collected from Kaggle by observing the dataset of credit card transactions with amount, time , transaction type with various features. The features are extracted with Principle Component Analysis algorithm.

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The next section discusses with Quantum machine learning algorithms.

IV. Quantum Machine Learning

A branch of quantum computing called quantum machine learning (QML) intends to use quantum algorithms and methodologies to address machine learning issues. The fundamental difference between quantum machine learning and classical machine learning is that the latter uses quantum mechanical effects such as superposition and entanglement to process information. In traditional machine learning, data is processed on classical computers using classical algorithms, whereas in QML, quantum algorithms operate on quantum data. Quantum bits, also known as qubits, which may exist in a superposition of states, can be used to represent quantum data. This means that a qubit can simultaneously represent multiple values, whereas classical bits can only represent one value at a time. Quantum Support Vector Machine (QSVM), is utilized for classification tasks, is one of the most significant quantum algorithms used in QML. Using a quantum kernel function, which is a quantum equivalent of a classical kernel function, input data is mapped by QSVM into a high-dimensional space.Quantum machine learning is used to handle large datasets which efficiently performs parallel computations which makes suitable for them to suitable in real-time fraud detection. Data is then separated in classes using a quantum version of the classical support vector machine (SVM) algorithm.

We looked at 3 methods for resolving this issue. They are Traditional methods such as Decision trees, KNN, Logistic Regression, SVM, Random forest trees, and XGBoost are examples of regression models. The fraud detection is achieved by implementing machine learning , deep learning or quantum machine learning algorithms. In the proposed work, the detection of attacks are implemented with SVM,Random forest, XGBoost and Quantum Neural Networks

This model's application can be found in classical ml.py. The next technique is Neural networks are used where three-layers are used strictly to conventional neural network. discoverable in classical nn.py. Next, Quantum hybrid which are neural network, where it is sandwiched with a quantum layer between two conventional layers. The file hybrid nn.py performs this. To achieve this, a Keras layer is created by transforming a PennyLane quantum layer. Using the KerasLayer class of the qnn package, we turn the QNode into a layer, the fundamental building component of Keras.The QNode can be composed of any set of devices, actions, and metrics that are legal in PennyLane. The QNode parameters must, however, also meet other requirements, such as having an argument named inputs. The model treats all additional inputs as trainable weights and requires that they be either arrays or tensors.[17]. In the first layer we collect the data from the dataset and preprocess and clean it. In the second layer we separate the data into training and testing, after which we extract the features from the transactions. The last layer uses our three modules namely, classical machine learning which has 6 algorithms, a neural network and hybrid quantum computing. We detect fraud cases in the dataset and calculate the accuracy for each of the modules. Our architecture consists of 4 layers which are depicted in Fig 1.Quantum Neural Network (QNN), is quantum alternative of a CNN, is another significant quantum technique utilized in QML. QNNs can be used for a variety of tasks, including classification, regression, and optimization. QNNs can also be used for quantum state preparation, which is a process of preparing a quantum state that represents the input data.

One of the major challenges in QML is the issue of quantum noise and error correction. Quantum computers are inherently noisy due to the effects of decoherence and other sources of error, and this can affect the accuracy of QML algorithms. Therefore, developing effective error-correction techniques and algorithms that are robust to noise is a critical area of research in QML. Overall, QML has the ability to completely change the machine learning industry by providing exponential speedups for certain tasks and enabling new approaches to data processing and analysis. However, much research is still needed to fully realize the potential of QML and to overcome the challenges associated with quantum noise and error correction. Now the detailed implementation is given with each layer.

A. DATA COLLECTION LAYER

The dataset comprises credit card purchases made in September 2013 by European cardholders. In this set of contacts, which included transactions data which have been recorded for 2 days, there were 492 frauds transactions out of 284,807 total transactions. In the total data scams are accounted for 0.172% of all the transactions that were put in the positive class, as the data is very unbalanced. Only data which could be presented numerical are there which have been modified using PCA (Principal Component Analysis) are present. As the confidentiality should be considered although we are using the real life data, so were not able to disclose baseline attributes and extra information about the data. The primary PCA components are features V1, V2,..., and V28. "Time" as well as "Amount" are characteristics which have not changed by PCA. "Time" component has a list of the seconds between each transaction and the first transaction in the dataset. PCA is used to reduce large dimensionality vectors when one is combining a large number of variables into a smaller set, retaining most of the information in the larger set. As a data set's variables are decreased, accuracy unavoidably declines; nonetheless, dimensionality reduction can be accomplished by sacrificing some accuracy for being simple. With smaller data sets, machine learning algorithms can analyses data much more quickly and easily since there are less irrelevant factors to take into account. Data pre-processing involves following steps like data collection, data cleaning ,data transformation ,data imbalance ,data splitting and data augmentation. Data cleaning is used to handle missing values such as mean, median or mode. For data transformation min-max scaling algorithm is used. One Hot encoding algorithm is used for feature engineering.

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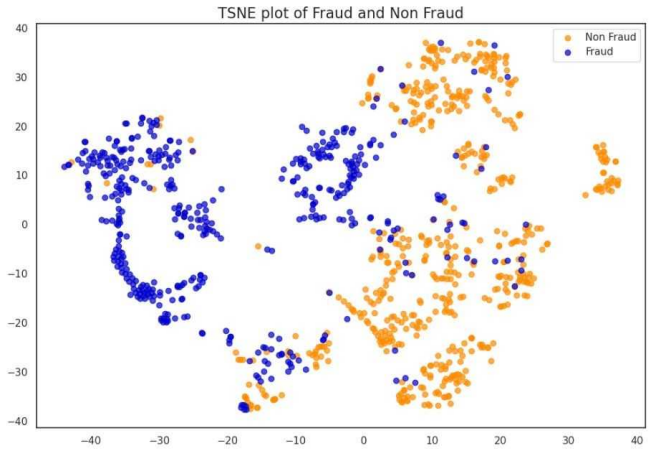

FIGURE2. t-SNE (t-Distributed Stochastic Neighbour Embedding) plot of Fraud and Non Fraud transactions

In Fig 2 A machine learning approach called t-SNE is used for data visualization and dimensionality reduction is depicted. Geoffrey Hinton and Laurens van der Maaten designed it in 2008. The technique seeks to preserve the similarity links between the data points while mapping high-dimensional data onto a lower-dimensional space (often two or three dimensions). It is particularly useful for visualizing complex, nonlinear structures in high-dimensional data. The next part discusses the algorithms which are used in the proposed model.

B. ALGORITHMS EXPLAINED

The algorithm SVM is used for effectively analyzing high dimensional spaces and less overfitting. The random forest algorithms are used to combine ensemble approach with decision trees and reduces overfitting. Further, the random forest algorithms are used to provide scalability with the data set. The XGBoost algorithm is used for providing sparse data with missing value and avoids overfitting. The quantum neural networks are used to enhance speedup, complex data, quantum superposition. F1 score is a metric which measures a model's accuracy. It combines the recall and precision of the model. The Decision trees are a mechanism for arranging the algorithm in machine learning. Using a cost function and a decision tree approach, the dataset's features will be separated. Before the decision tree is optimised, pruning, a technique used to eliminate branches that could use unnecessary characteristics, is performed. The Accuracy is 99.95 and F1 score is 83.01. It then makes use of the KNN model, which actually involves a type of supervised machine learning technique dealing with regression and classification prediction problems. It is mostly used in industry to solve forecast and classification problems. The results for this technique will include an F1 score of 88 and an accuracy of 99.966. The next technique in machine learning goes as logistic regression, making it one of the most popular among all supervised learning techniques. It is a method of predicting the categorical dependent variable based on a given set of independent factors or variables. The F1 score is 73.75 and the accuracy is 99.93. The next technique is called SVM. It is a set of supervised learning techniques known as support vector machines (SVMs) used in data categorization, regression analysis, and anomaly detection. A plus to support vector machines includes their high efficiency in high-dimensional situations. However, this is useful when there are more variables than data. The F1 score is 83.91 and the accuracy is 99.95. Followed by a machine that learns on a random forest tree. One of the components within guided learning is a random forest. It can be used not only in Machine Learning to solve problems related to regression and classification but also for the concept of ensemble learning. This is where several classifiers are combined to help solve complex problems and improve models. This approach has an F1 score of 80.55 with an accuracy of 99.94 %. It starts with XGBoost, a distributed gradient-boosting program that is impressively efficient, flexible, and portable due to optimization.

The technique employs Gradient Boosting to execute machine learning techniques. A variety of data science issues may be successfully resolved by using the parallel tree-boosting method known as XGBoost, sometimes referred to as GBDT or GBM. The algorithm Accuracy is 99.964, and F1 score is 87.24. Next the Neural Network uses a technique that mimics how the human brain functions, a neural network is a group of algorithms that seeks to find hidden connections in a piece of data. Neural networks in this context are systems of neurons that can have either an organic or synthetic origin. Without modifying the output criterion, neural networks can give the greatest results because they can adapt to changing input Without modifying the output criterion, neural networks can give the greatest results because they can adapt to changing input. In Fig 3 the Accuracy is 99.95 depicted with training and testing data.

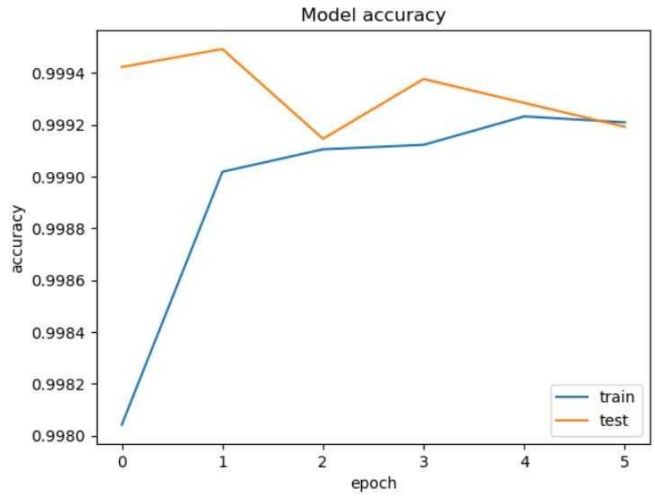

FIGURE 3. Neural Network model accuracy graph

Next , the pennylane technique is used to implement PennyLane enables users to train quantum circuits by merging machine learning frameworks with quantum simulators and hardware. Utilizing common scientific computing or machine learning libraries, such as SciPy in Python, conventional calculations, model optimization, or model training are carried out. In order to provide a bridge between classical and quantum computing, PennyLane combines these libraries with quantum simulators. The accuracy achieved is 99.99%, which is depicted in Fig 4.

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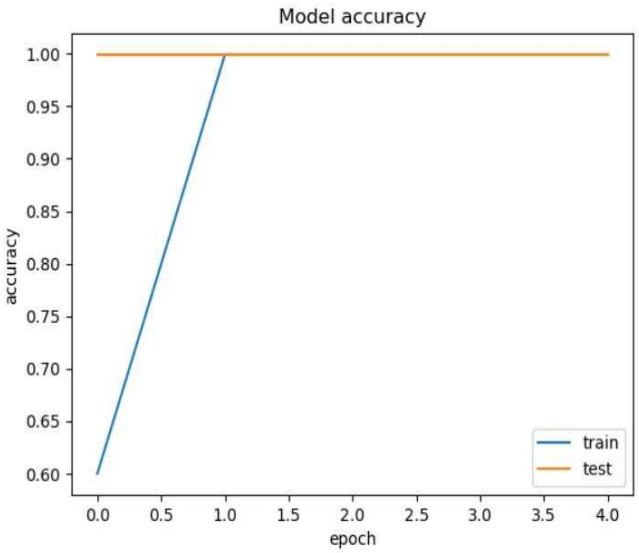

FIG 4. Quantum hybrid model accuracy grap

V. RESULTS AND DISCUSSION

For the output of the traditional ML methods, when it comes to precision, F1, and memory, the K-NN algorithm excels. The below Fig 1 displays the comparisons of classical approach is discussed in detail. The accuracy in the proposed work is achieved by implementing high quality data from the dataset and feature engineering approaches. Hence in the proposed dataset the accuracy is achieved by properly implementing quantum neural network with hyperparameter tuning. The SVM algorithm consists of kernel functions which are used to handle non-linear relationship and maximizes the clear decision boundary in classification task and it is memory efficiency. The random forest algorithm consists of ensemble of decision trees. The Quantum Neural network provides proper hyperparameter tuning which produces high accuracy for the model. The precision of the proposed algorithm is improved by using algorithms such as XG Boost and random forest algorithms. The detection rate of random forest algorithms is 90% for the dataset, then SVM provides the detection rate as 85%, Quantum Neural Network provides detection rate as 92% for the dataset.

Results regressive machine learning algorithms (no SMOTE)			
	accuracy	F1 score	recall
Decision tree	0.999509	0.830189	82.50%
KNN	0.999667	0.88	82.50%
Logistic Regression	0.999316	0.737588	65.00%
SVM	0.999574	0.839161	75.00%
Random forest tree	0.999482	0.805556	72.50%
XGBoost	0.999648	0.872483	81.25%

FIG 5. Model Comparison for classical approach

VI. CONCLUSION

The findings attained to the classical neural network were in a comparable success range to those of the classical regressive methods used in the above image. Although we could only execute the hybrid algorithm on 10% of the data set due to lengthy calculation periods, early results of the algorithm indicated it was functioning properly. Only 10% of the entire data collection was used for the hybrid neural network's initial training outcomes. Though the method appears to be functioning properly, the processing times are excessive. In comparison to its traditional counterpart, the hybrid neural network's early data indicates longer computational periods. However, it exhibits quicker accuracy convergence when performed for equivalent quantities of data points, batches, and epochs. The performance of the proposed work in future can be improved by integrating ensemble machine learning algorithm with Quantum machine learning algorithms.

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